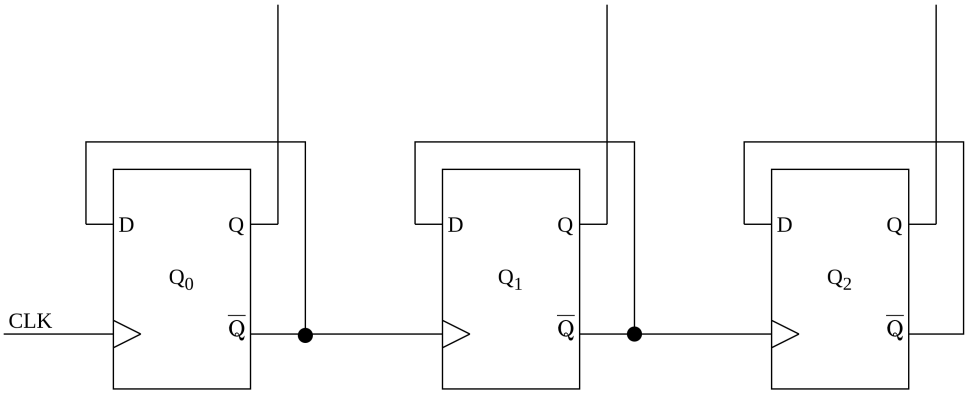
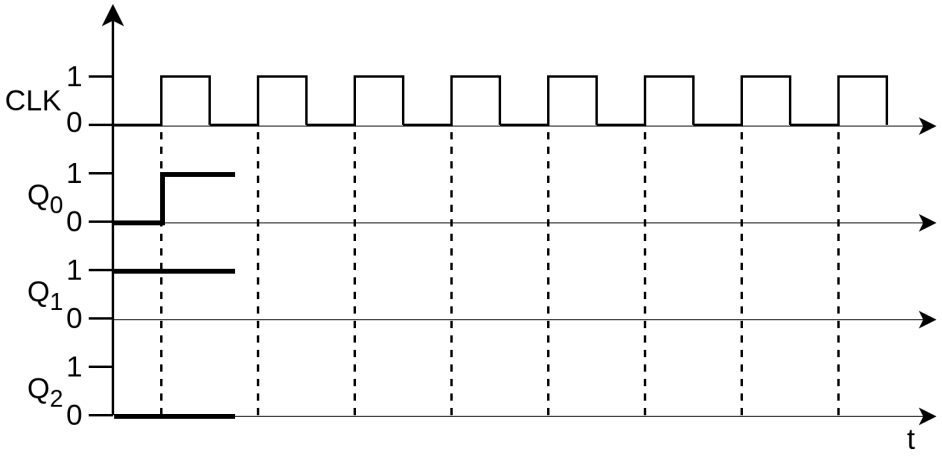
Sequential circuits

A 3-bit ripple counter is shown below:



(a) Relating to a rising edge on the `CLK` signal, describe exactly how this could cause a change of state of `Q1`.

(b) On the axes below, complete the waveforms for Q0, Q1, Q2, and Q3 given the CLK input to the 3-bit ripple counter

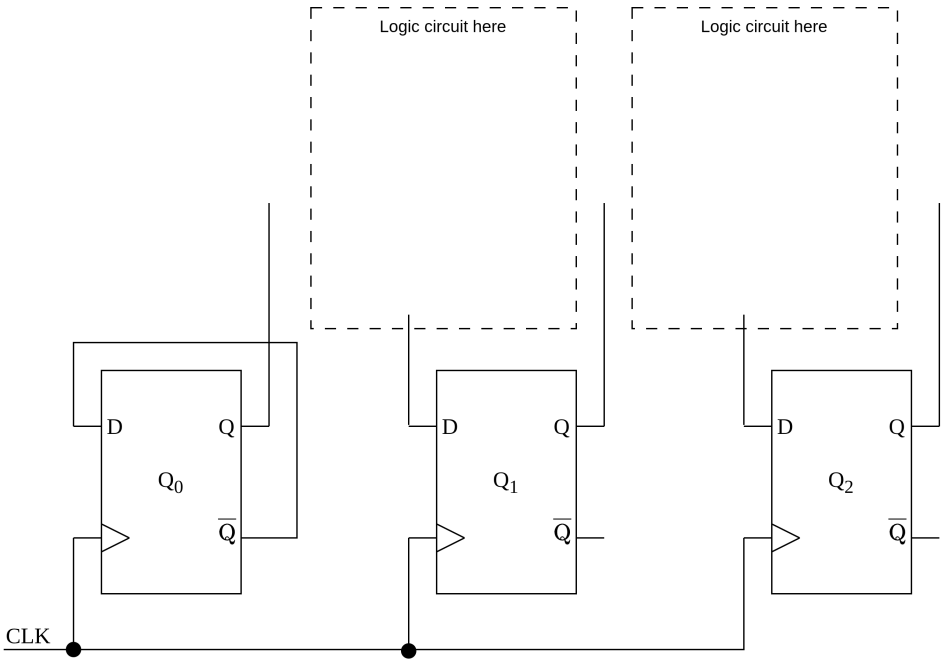


(c) Considering the state change from `111` → `000`, explain why all bits do not change at the same time. In your answer, state whether the circuit is synchronous or asynchronous, and explain any intermediary states.

(d) Why is the behaviour explained in part (c) undesirable?

(e) We are going to fix the behaviour explained in part (c) by the addition of some combinational logic circuits, which will determine the next state of each bit for a 3-bit counter. Given the table below, complete the combinational logic design for $Q_{1,i+1}$ and $Q_{2,i+1}$, and draw the logic circuits in below.($Q_{0,i+1}$ is done with $\overline{Q_{0,i)}}$)

Current			Next		
$Q_{2,i}$	$Q_{1,i}$	$Q_{0,i}$	$Q_{2,i+1}$	$Q_{1,i+1}$	$Q_{0,i+1}$
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0



(f) For your solution, explain how you have solved the problem of asynchrony (i.e. explain when all of the flip flops change state from `111`, and what value is present on each flip flop's input, `D`). You can play with an equivalent circuit here.

Lecture 5 - CPU

(a) Write an expression for a full adder's C_{out} in terms of the inputs A, B, and C_{in}.

(b) Draw a circuit using AND, OR, and NOT logic gates that implements your expression for C_{out}.

Lecture 6 - Combinational circuits

(a) What would the logic circuit for an 8:3 encoder look like? You may use 4-way `OR` gates in your answer.

(b) Implement a 4:2 mux using two 2:1 muxes.

(c) Carry out the design process from a 4:1 multiplexer. You may use 3-way `AND` gates in your answer.

GPIO

Figure 1 is a simplified circuit diagram for a GPIO pin on the ATmega32U2 microcontroller.

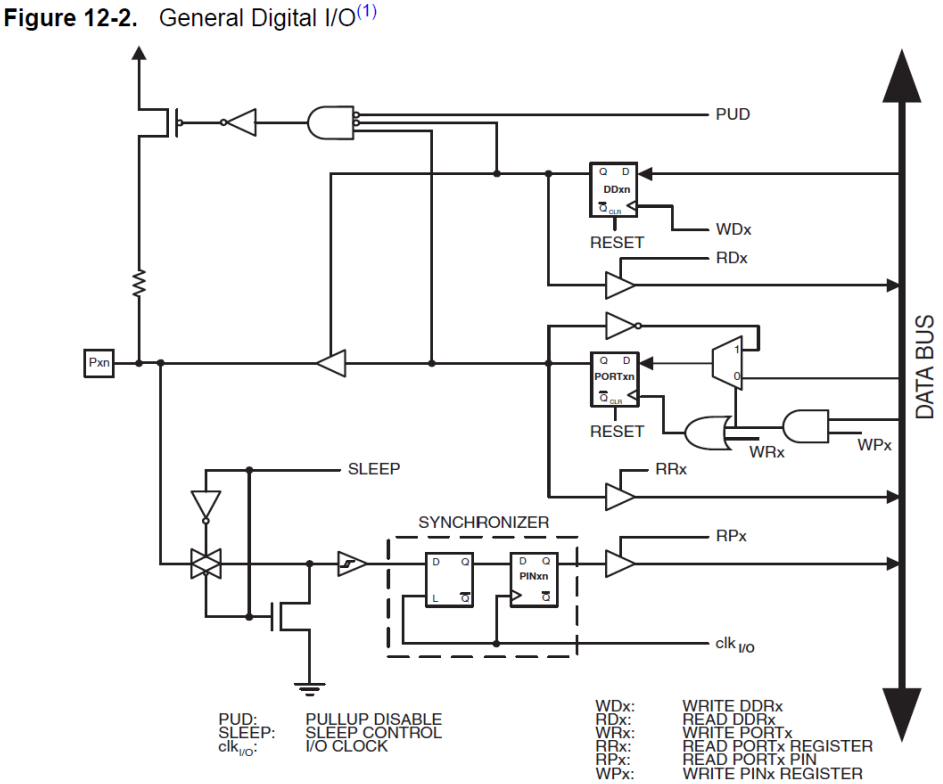


Figure 1: The logical circuit for one GPIO pin on the ATmega32U2 MCU.

(a) The `PUD` signal is a "Pull-Up Disable". Briefly explain why setting PUD to 1 would disable the pull-up resistor connected to *Pxn*.

(b) What is the boolean expression for enabling the pull-up resistor?

(c) Under what conditions would the tristate buffer between *Pxn* and the *PORTxn* flip-flop be turned ON, i.e. conducting?

(d) What values need to stored in the DDxn and PORTxn flip flops to output a digital `HIGH` signal on the pin?

(e) (Extra) What is the role played by the 2:1 multiplexer that provides the input to the *PORTxn* flip-flop?

Learning about computers using computers

Complete up to the multi-bit adder in the Nandgame (https://nandgame.com/).

END OF QUESTIONS